MIL-P-36174A (DSA-DM)
11 August 1970
SUPERSEDING
MIL-P-36174 (DM)
12 December 1963

MILITARY SPECIFICATION

PILLOW, PNEUMATIC

This limited coordination Military Specification has been prepared by the Defense Personnel Support Center based upon currently available technical information, but has not been approved for promulgation as a coordinated revision of Military Specification MIL-P-36174 (DM). It is subject to modification However, pending its promulgation as a coordinated Military Specification, it may be used in procurement.

1. SCOPE

1.1 This specification covers one type of pneumatic pillow, suitable for use on aluminum folding cots in field military hospitals.

2 APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

MMM-A-250	Adhesive, Water Resistant (For Sealing Waterproofed Paper)
PPP-B-566	Boxes, Folding, Paperboard
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard
PPP-B-676	Box, Setup.
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, (for Carton Sealing)

FSC 7210

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Military

MIL-L-10547 Liners, Case, and Sheet, Overwrap, Water-Vaporproof or Waterproof, Flexible.

STANDARDS

Federal

Fed. Std. 191 Textile Test Methods.

Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 Marking for Shipment and Storage.

(Copies of specifications and standards, required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer)

3 REQUIREMENTS

3.1 Material

3.1.1 Basic fabric. The basic fabric shall be 36-inch 4.0 cotton sheeting which shall be constructed and processed to have the following minimum characteristics

Yarn count	Warp 56 Filling 53
Breaking strength	Warp 50 pounds Filling 45 pounds
Weight per square yard	3.5 ounces
Color	Olive drab or khaki

3.1.1.1 Coating. The basic fabric shall be rubber coated as follows

2 spreader coats	Total weight 3/4 ounce per square yard
2 calendered coats	Total weight 13-1/2 ounces per square yard

The total weight of the finished fabric shall be 18 ounces per square yard

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3.1.1.2 Hinge and stay material. The hinge and stay material shall conform to 3.1 1.1 except that the cloth need not be dyed and the total weight of the coating shall be five (5) ounces per square yard minimum.

3.1.2 Cement. The compound shall be a synthetic or natural rubber cement, completely compatible with the coating on the fabric.

3 2 Construction. The pillow shall be constructed from one-piece top and bottom panels of the coated fabric specified in 3.1.1.1, assembled with the coated side of the fabric on the inside of the air chambers. No patches shall be permitted. The top and bottom panels shall be cemented or molded together forming a fin edge 3/8 inch ± 1/8 inch in width. When the fin edge is cemented together, a strip of the hinge material specified in 3.1.1.2, 1-1/2 inches ± 1/8 inch wide, folded as a hinge, shall be fully cemented to the top and bottm panels on the inside along the entire perimeter of the pillow The pillow shall have three air chambers formed by use of one 15 inch + 0. -1 inch long and one 8 inch + 0, -1 inch long seams which can be either pressure molded (all in one piece) or cemented. When cemented construction is used, these seams shall be reinforced with 2-1/2 inch wide stays or risers, using material as specified in 3 1.1 2. The long seam shall run parallel to the long dimension and the short seam shall be parallel to the short dimension midway between the two edges, thereby forming the three air chambers. The valve (see 6.1) shall be airtight sealed into one corner of the pillow (one of the two small air chambers) as follows:

A 3-inch square piece of coated fabric (see 3.1.1.1), with the valve positioned in its center, shall be doubled over to form a triangle 3 inches x 3 inches x 4-1/4 inches The triangular sections (with valve) shall be cemented between the top and bottom panels at one corner so that the valve remains completely covered. A rubber coated reinforcement tape may be used to make the valve airtight. The uninflated pillow shall measure 16 inches ± 1/2 inch by 17-3/4 inches ± 1/2 inch long The inflated pillow shall measure 13 inches ± 1/2 inch width by 16-1/2 inches ± 1/2 inch in length.

3 3 Identification marking Each pillow shall be marked with the manufacturer's name or registered trademark Markings shall be stenciled or printed with waterproof stencil paint or waterproof ink, and shall be in a conspicuous location. Markings shall be of appropriate size and distinct, legible, and of a permanent nature.

3.4 Workmanship Workmanship shall be first class throughout Pillows shall be free from defects which detract from their appearance or may impair their serviceability

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4. QUALITY ASSURANCE PROVISIONS

* 4.1 Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4 2 Pre-Acceptance Inspection.

4.2.1 Preproductior Inspection. When required, inspection and testing of the preproduction sample shall be made of a completely fabricated pillow for all provisions of this specification applicable to end product examination and tests.

4.3 Inspection for Acceptance. Unless otherwise indicated, inspectior shall be in accordance with MIL-STD-105

4.3 1 Inspectior of Components.

4.3 1.1 Component Testing In addition to the quality assurance provisions of the subsidiary specifications, testing shall be performed on components and materials listed ir Table I for the test characteristics shown.

4.3.2 Certification of Compliance Components and materials listed below ma' be acceptec or the basis of a contractors certification of compliance for characteristics indicated.

Component	Characteristics	Rqmt Para
Coated fabric	Fabric coated with 2 spreader coats (total wt of 3/4 ounce/sc yd) and 2 calendered coats (total wt of 13-1/2 ounces/sq. yd)	3.1.1.1
Hinge anc Stay Material	Same as for 3.1.1.1 except that total weight of coating 5 ounces/sq yd. min.	3.1.1.2
Cement	Compound used for cementing shall be synthetic or natural rubber cement, completely compatible with the coating on the fabric.	3.1.2

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TABLE I

COMPONENT	LOT EXPRESSED IN TERMS OF	CHARACTERISTIC	REQMT PARA	TEST METHOD	REQUIREMENTS APPLY TO SPL UNIT	REQUIREMENTS APPLY TO LOT AVG	# DET PER SPL UNIT	S O RESULTS REPORTED AS	INSP LV	AQL
Base fabric prior to coating	1 yard	Yarn count Warp	3.1.1	Fed Std 191 5050	X		5	1/2 yard	II	6.5
		Yarn count Filling			X		5			
		Breaking Strength Warp		5100	X		5	Avg of 5 det to nearest unit	II	6.5
		Breaking Strength Filling			X		5			
		Weight		5041	X		1	Avg of 5 det to nearest 1 lb	II	6.5
Coated fabric	1 yard	Weight	3.1.1.1	5041	X		5	1/2 yard Avg of 5 det to nearest 0.1 oz/sq yd	II	6.5
		Color		Visual		X	1	Pass or fail	II	

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4.3 3 In-Process Inspection. Inspection shall be made at any point or during any phase of the manufacturing process to determine whether operation or assemblies are accomplished as specified. The Government reserves the right to exclude from consideration for acceptance any material for which in-process inspection has indicated non-conformance In-process inspection shall be conducted to see that accomplishment of the following is in accordance with the specification requirements

Requirement, operation or assembly	Requirement
Hinge material properly cemented around entire inside perimeter of the pillow (when required).	3.2
That 15 inch and 8 inch long stays are used to reinforce the seams between air chambers.	3.2

4.3.4 End Item Examination. The defects found during examination shall be classified in accordance with 4 3 4 1, 4.3 4 2 and 4 3.4 3 The lots shall be inspected in accordance with 4 3.4.4 The sample unit for these examinations shall be one completely fabricated pneumatic pillow

* 4.3 4.1 AQL s and Inspection Levels. The inspection levels and acceptable quality levels (AQL's) expressed in defects per hundred units shall be as follows

	Inspection level	AQL's Major	AQL's Total
For defects applicable to 4.3 4 1	I	2 5	6 5
For defects applicable to 4.3 4.2	S-3	One class	4 0
For defects applicable to 4 3.4 3	S-3	One class	4 0

4.3.4.2 Examination for Visual Characteristics Examination of the pillow for defects in material, design, construction, workmanship and marking as shown in Table II

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Table II

	EXAMINE	DEFECT	MAJOR	MINOR
	Quality of coated fabric	Not fabricated from specified Material	X	
	Material	Hole, cut, tear or weak place	X	
	Finish	Any visible deterioration	X	
		Shade variations between panels affecting appearance		X
		Fabric not soft and pliable		X
		Color and shade not as required		X
	Odor	Rancid or objectionable	X	
	Spots or stains	Fabric dirty or greasy to an extent that it cannot be completely removed with a damp cloth	X	
		Clearly noticeable when viewed at a distance of 6 feet		X
*	Design and Construction	Pillow not vulcanized all in one piece	X	
		Top or bottom not 1 piece	X	
		Any component missing	X	
		Patches on pillow	X	
*	Fin edge	Missing	X	
		Not cemented or molded along entire perimeter of pillow	X	
		Any open or unfilled channel in fin edge	X	
		Uneven trim (with some areas under minus tolerance)	X	
		Uneven trim (with some areas over plus tolerance)		X

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4.3.4.1 Examination for Visual Characteristics (Cont'd)

EXAMINE	DEFECT	MAJOR	MINOR
Seams	Missing, loose or broken	X	
	Misplaced seriously affecting serviceability or appearance	X	
	Misplaced, not seriously affecting serviceability or appearance		X
	Not pressure vulcanized along entire length	X	
Valve	Not specified type	X	
	Not located as required Seriously affecting serviceability	X	
	Not seriously affecting service-ability		X
	Valve not air tight sealed	X	
	Valve stopper does not function properly	X	
Workmanship	Any adhesion between top and bottom panels (except where required)	X	
	Any sharp crease or wrinkle affecting appearance		X
Marking	Missing, incorrect, illegible		X
	Not accomplished in specified manner		X
	Not in conspicuous location		X

* 4.3.4.1 Examination of uninflated Pillow for Dimensions Any dimension that is not within the specified tolerances shall be classified as a defect

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4.3.4.4 Examination for Inflation Characteristics. Examination of the pillow for defects in the inflated condition.

EXAMINE	DEFECT
Inflated pillow	Evidence of leakage (not air tight)
	Not easily inflatable by mouth or pump
	Blocking resulting in closure and lack of inflation of any chamber

* 4.3.5 Examination of Preparation for Delivery Requirements. An examination shall be made to determine that applicable packaging, packing and marking requirements are complied with. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be sealed Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot The inspection level shall be S-2 and the AQL shall be 4.0 defects per hundred units

Examine	Defect
Marking (Exterior and Interior)	Omitted, incorrect, illegible, of improper size, location, sequence or method of application
Materials	Any component missing
	Any component damaged, affecting serviceability
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, inadequate stapling
	Bulging or distortion of containers
Weight or content (Exterior and Interior)	Number of end items is more or less than required, gross/net weight exceeds requirements

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5 PREPARATION FOR DELIVERY

5.1 Packaging Packaging shall be level A or C as specified (see 6.2)

5.1.1 Unit of issue One pillow, as specified, constitutes one unit of issue

5.1.2 Packaging quantities. The number of units of issue indicated in Table III shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table III Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	6 units	72 units

* 5.1.2.1 Packing variation permitted If the required number of units in the entire shipment is less than the number of units specified to be over-packed in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A

* 5.1.3.1 Unit package Unit package shall be a box of appropriate size and design constructed in accordance with PPP-B-566 or PPP-B-676, except commercial colors will be acceptable, or PPP-B-636, type CF, class domestic Closure shall be adequate to prevent accidental opening under normal handling

* 5.1.3.2 Intermediate package Intermediate package shall be a box of appropriate size and design constructed in accordance with PPP-B-636, type CF, class weather-resistant Boxes shall be closed by taping the top and bottom flaps with 2 inch wide tape conforming to PPP-T-76 Closing of bottom flap with adhesive conforming to MMM-A-250 or metal stitches is permitted

5.1.4 Level C Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

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5.2 Packing. Packing shall be level A, B or C as specified. (See 6 2).

5.2 1 Level A

* 5 2 1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3, PPP-B-601, overseas type, PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3C Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.2 Level B

* 5 2.2 1 Exterior container Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type, PPP-B-621, class 1, or PPP-B-636, class domestic Closure shall be as specified in the appendix of the applicable box specification

5 2 3 Level C The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5 3 Marking

5 3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129 When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material Contract or purchase order number and name of contractor shall be shown

* 5 3 2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129

6 NOTES

* 6 1 The valves referred to in 3.2 are available as

Parker and Harper No 61C or
Dill No 8630A,

Schrader No 9933, or
Halkey Roberts No. 26C AS

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6.2 Ordering data. Procurement documents should specify the following

(a) Title, number and date of this specification
(b) Selection of applicable levels of packaging and packing (see Section 5).

6.3 This specification does not include all types, classes, grades, and sizes, of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, which are normally procured to meet Military requirements.

* 6.4 This specification covers the following item appearing in the Federal Supply Catalog.

FSN	Item
7210-299-8520	Pillow, Pneumatic

6.5 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue

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INSTRUCTIONS This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost Comments and the return of this form will be appreciated Fold on lines on reverse side, staple in corner, and send to preparing activity Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document (s) or serve to amend contractual requirements

SPECIFICATION

MIL-P-36174A(DSA-DP) PSN:

ORGANIZATION

CITY AND STATE	CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO (If "yes" in what way?)

4 REMARKS (Attach any pertinent data which may be of use in improving this specification If there are additional papers attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity - Optional)	DATE

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